Social Network Analysis (SNA)

B Tech III CSE

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Week #2 <Holiday> <Extra Lecture rescheduled on 22 Jan 2026>

The Greenland Saga

Greenland is an autonomous territory of the Kingdom of Denmark

Why Greenland matters

Arctic gateway: unlock new shipping routes, potentially reducing Asia–Europe transit times by up to 40% as ice recedes.
Critical resources: The island is rich in rare-earth elements - all vital for advanced technology and defense industries.
Military significance: The U.S. already operates the Pituffik (formerly Thule) Space Base, a cornerstone of missile-warning and space-surveillance systems.

Where allies stand Denmark and Greenland's leadership have firmly rejected any idea of a sale, stressing Greenland's right to self-determination. European allies—including France, Germany, Norway, and Sweden—have made symbolic military deployments to signal NATO solidarity and deter escalation.
Russia's position Officially, the Kremlin maintains that Greenland is Danish territory.

Back to business
Where We Are Coming From

- Networks shape power, risk, influence
- Structure matters more than individuals
- Cascades and coordination
- This week: formalizing intuition

From Stories to Models

- You learned formulas with Dr AC and implemented metrics in lab
- Today, we move from intuition to rigor
- Graphs provide computable structure
- SNA becomes a formal analytical science
- **From Metrics to Strategic Judgment in Social Network Analysis**

Revisiting: Network as Graph

- **A social network = Graph**
- Nodes: (Actors, entities)
- Edges: (Relationships, interactions)
- Abstraction enables modelling
- Nodes Edges Examples:
- Students ↔ friendships
- Devices ↔ communication
- Accounts ↔ message forwarding
- Countries ↔ alliances

What Can Be a Node?

- People, devices, organizations
- Countries, documents, concepts
- E.g. Research Papers (citation networks)
- E.g. Concepts (knowledge graphs) –
- A Knowledge Graph (KG) is a network where nodes are concepts (or entities) and edges are semantic relationships
- Node definition depends on analytical question

Example: Simple Concept Network

- Let us take the domain of cybersecurity.
- Nodes (concepts): Malware, Phishing, Zero Trust, Authentication, Identity, Trust etc.
- Edges (relationships):
- Phishing → *is a type of* → Social Engineering
- Authentication → *supports* → Zero Trust
- Identity → *is foundational to* → Trust
- Malware → *exploits* → Vulnerability
- This forms a **knowledge graph of the domain**.

More on Knowledge Graphs

- Paths represent chains of reasoning
- Central nodes represent core concepts
- Bridges represent interdisciplinary ideas
- Clusters represent subfields
- Now SNA-style questions are:
- Which concepts are most central in cybersecurity?
- Which ideas bridge AI and security?
- Which concepts dominate the curriculum?
- Where are the knowledge gaps?

Graphs

Graph Type	**Nodes Represent**	**Edges Represent**
Social Network	People	Relationships
Communication Graph	Devices	Messages
Knowledge Graph	Concepts	Meaningful relations
Citation Graph	Papers	Citations
Dependency Graph	Modules	Dependencies

Metric Failure Modes

- **A. When Degree Centrality Fails**
- Example:
- Celebrity with 10M followers but no trust
- Bot account with high degree
- Corporate org chart: manager vs real influencer
- **B. When Betweenness Is Misleading**
- Betweenness depends on shortest-path assumption
- Real-world flows don’t always follow geodesics
- Dynamic networks change centrality rapidly
- **C. When Closeness Breaks**
- Disconnected components
- Directed networks with unreachable nodes

Takeaway

- Social networks explain how influence flows between people.
- Knowledge graphs explain how meaning flows between ideas.

Edges Are Not All Equal

- Directed or undirected
- E.g. Directed (A → B follows on Twitter)
- E.g. Undirected (Friendship on Facebook)
- Weighted or unweighted
- E.g. Weighted (Strength of interaction)
- Static or temporal
- E.g. Temporal (Edges appear/disappear)
- Trust-based
- Adversarial

Types of Networks

- Social, communication, collaboration
- Biological, infrastructure,
- Security and cyber ecosystems
- E.g. Terror networks, Botnets
- Supply chains, Defence ecosystems

Real Networks Are Structured

- Clusters and communities
- Hubs and bridges
- Small-world effects

Properties observed in real networks:

- Clustering (communities)
- Hubs (high-degree nodes)
- Small-world effect //next slide
- Scale-free structure

These patterns explain why cascades occur.

SWE (Small-world effect)

- A network exhibits the **small-world effect** if it simultaneously has:
- **High clustering coefficient** (nodes' neighbors are also connected to each other)
- **Low average shortest path length** (any node can be reached from any other via very few steps)
- In plain terms:
- Networks feel locally dense but globally small.

SWE - example

- You are closely connected to:
- Your friends
- Their friends
- Their friends' friends
- Yet, surprisingly, you are only:
- ~5–6 steps away from almost any human on the planet.
- This is popularly known as:
- **"Six Degrees of Separation"**

SWE - Structural Consequences

Small-world networks enable:

- Extremely fast information diffusion
- Rapid spread of rumors and misinformation
- Fast contagion (biological or digital)
- Efficient coordination (good or bad)

This directly explains why:

- WhatsApp rumors escalate rapidly
- Viral content explodes unexpectedly
- Movements (social or political) mobilize quickly

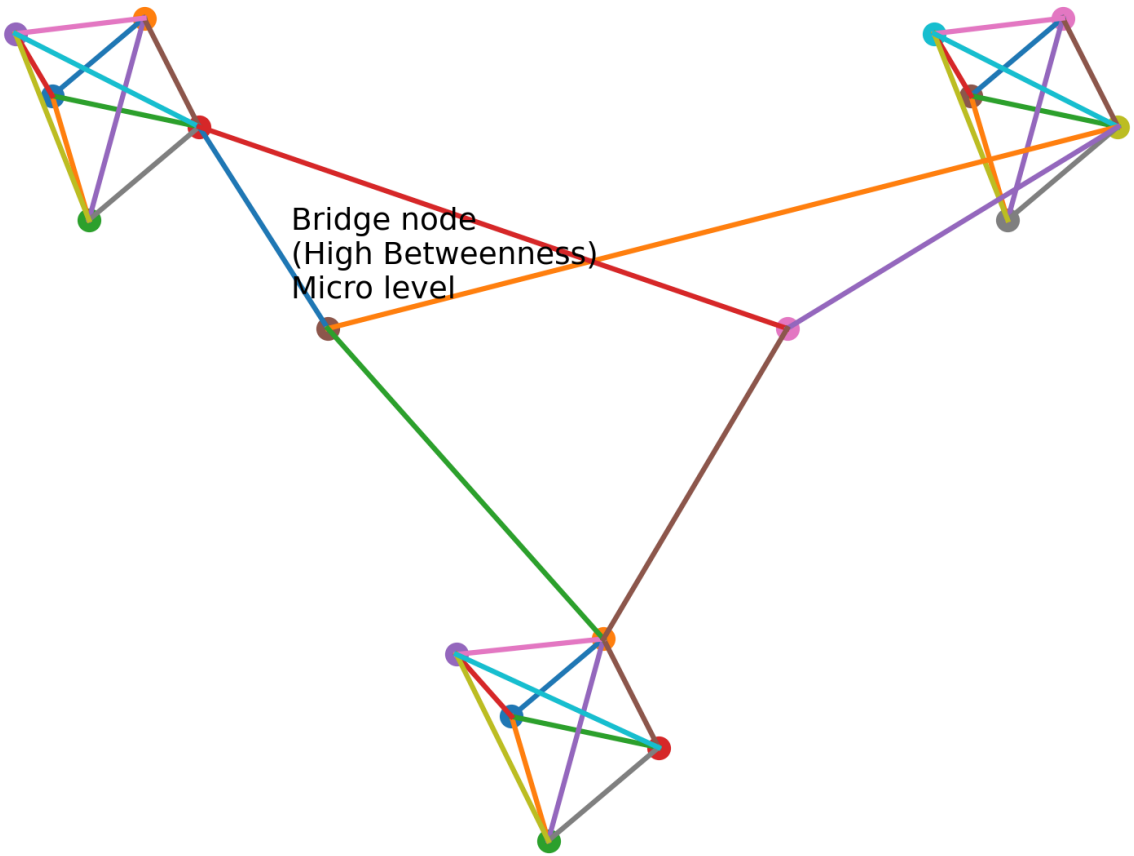
SWE Summary

- Two key metrics:
- **Clustering coefficient (C)** → High
- **Average path length (L)** → Low (grows logarithmically with network size)
- A network is small-world if:
- My friends know each other (high clustering)
- I can reach almost anyone through very few hops
- This explains the speed of:
- Virality
- Panic
- Collective action

Community 1 (Meso)

Community 2 (Meso)

Micro, Meso, and Macro Perspectives in One Network



Community 3 (Meso)

Macro level: Overall network structure
(dense clusters + sparse bridges)

Extra class on 22 Jan 2026

- Last class we built intuition about structure. Today we focus on judgment, failure modes, and application

Scale-Free Structure

- A network is **scale-free** when:
- The distribution of node degrees follows a **power law**
- Meaning:
- Most nodes have very few connections
- A tiny number of nodes (hubs) have extremely many connections
- Small-world explains **speed of spread**.
- Scale-free Structure explains **inequality of power**.

Example SFS

- Think of Airports:
  - Many small airports
  - Few mega-hubs (Delhi, Heathrow)
- Social media:
  - Millions with 10–100 followers
  - Few with millions of followers
- The network is **naturally unequal**.
- Interesting observation
- Growth – networks expand over time
- Preferential attachment – “the rich get richer”
- Nodes already popular attract new connections faster

SFS

- Scale-free structure explains: **Robustness vs Fragility**
- Random node failures → network survives
- Targeted attacks on hubs → network collapses
- This is why:
- Removing random social media users does nothing
- Removing key influencers destabilizes movements
- Taking down botnet controllers cripples the network
- Attacking major routers disrupts the internet

Summary SFS

- In scale-free networks: Few nodes matter enormously, Most nodes are structurally irrelevant
- Power, influence, and vulnerability are structurally concentrated.
- **Strategic Implications f**or:
- Cybersecurity
- Counter-terrorism
- Platform governance
- Defence networks
- The question becomes:
- Which hubs must be protected?
- Which hubs must be neutralized?
- This transforms SNA into a **strategic instrument**, not just an analytical one.

Degree Centrality

- Number of connections
- High degree ≠ most important
- Position matters
- A high-degree node is:
- Socially active, Frequently contacted, Highly visible, Popular or busy
- But **high degree ≠ high power**.
- Degree = volume; Influence = structure
- E.g. A college WhatsApp group admin might message everyone, but the real influence may lie with a quieter student who connects multiple groups.

Strengths of Degree Centrality

- Degree is very useful when:
- Modeling **broadcast behavior**
- Detecting **super-spreaders** in epidemics
- Identifying **high-activity accounts**
- Spotting **spam / bot-like nodes**
- Understanding **local visibility**
- It answers: “Who talks to many people?”
- **Weaknesses** - Degree centrality fails when:
- A node has many links inside a **single cluster only**
- A node is noisy but not trusted

Why? not

- Covert networks (terrorist, criminal, espionage, etc.) often:
- Intentionally **minimize degree**
- Use **compartmentalization**
- Rely on **brokers** between cells
- Avoid centralized hubs
- Why?
- Because high-degree nodes are easy to detect.

Betweenness Centrality

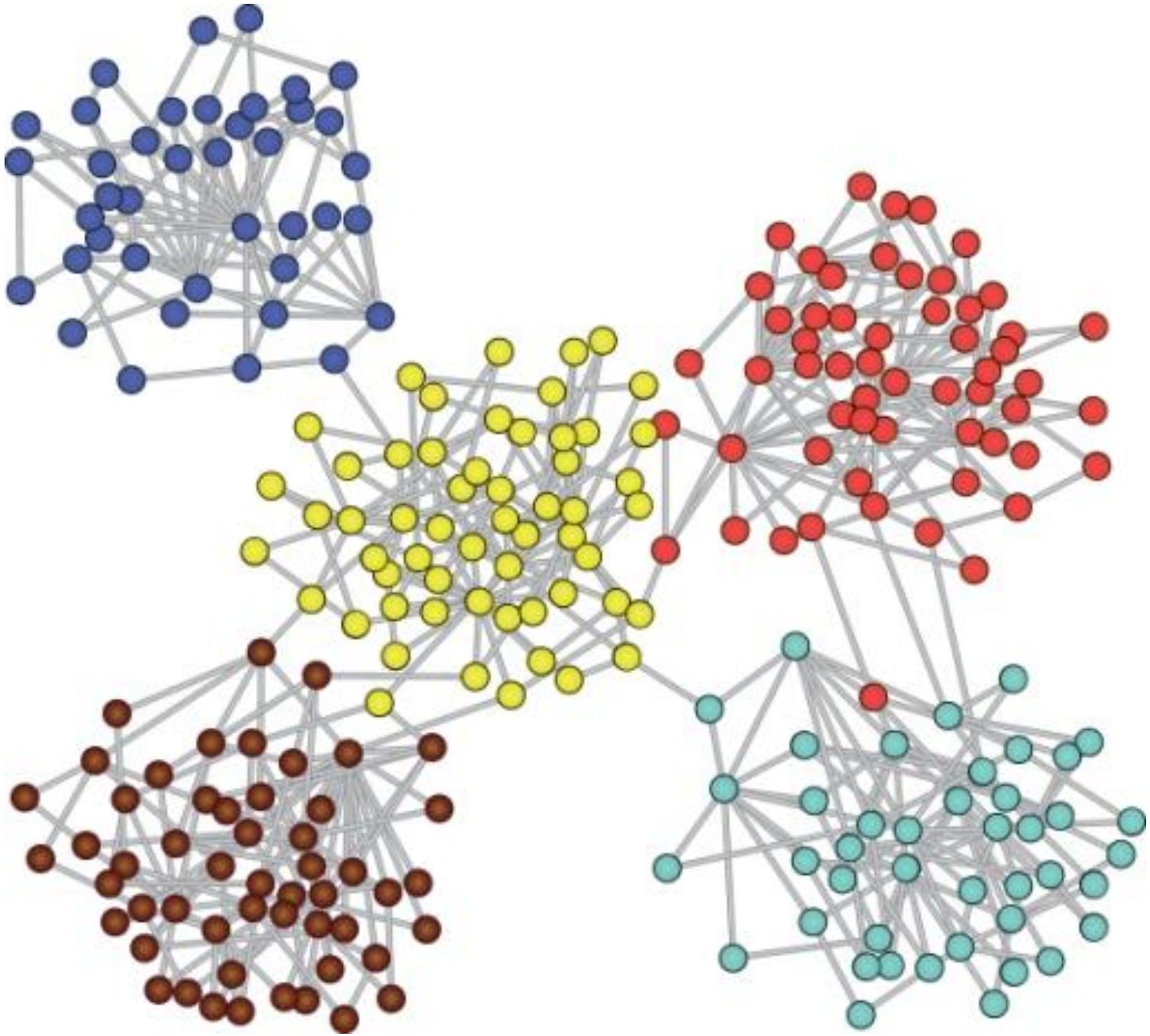
- Betweenness Centrality — The Power of Being "In Between"
- Betweenness measures **how often a node lies on the critical paths connecting others**
- Nodes that bridge groups
- Control information flow
- Strategic importance
- Degree is popularity. Betweenness is control.

Why Betweenness Matters More Than Popularity

- High-degree nodes are visible
- High-betweenness nodes are structural
- Real power often lies in:
  - Brokers
  - Intermediaries
  - Bridges
- Covert systems often hide leadership by minimizing degree
- Influence is often **invisible** unless you analyze structure.

How Betweenness Centrality Connects to Community Detection

- **Communities** are groups of nodes that are
- densely connected internally and
- sparsely connected externally
- If communities exist, then:
- Most paths **inside a community** stay inside
- Paths **between communities** must pass through a small number of nodes
- Those nodes are precisely the ones with:
- **High betweenness centrality**



Summary

- Degree centrality → tells you who is locally popular
- Betweenness centrality → tells you who connects groups
- Community detection → identifies what those groups are
- Degree tells you who is active.
- Betweenness tells you who is powerful.
- Community detection tells you where the real structure lies.

When Degree Centrality Misleads

- **Failure Mode 1: Popularity ≠ Power**
- High degree measures **visibility**, not **influence**
- Failure cases:
  - Bots / spam accounts with high activity
  - Couriers deliberately exposed in covert networks
  - Middle managers with many interactions but no decision authority
- Structural reality:
  - Power often resides in **low-degree but high-trust nodes**
- Analyst takeaway:
- Degree is a volume metric. Power is a positional property.

When Betweenness Centrality Lies

- **Failure Mode 2: Betweenness Assumes Unrealistic Flow**
- Betweenness assumes:
  - Information flows along **shortest paths**
  - Actors have **global routing awareness**
- In reality:
  - Humans follow trust, habit, and protocol—not geodesics
  - Organizational flows often follow **hierarchy**, not shortest path
  - Encrypted or compartmentalized networks distort paths
- Consequence:
  - Some high-betweenness nodes are mathematically central but operationally irrelevant
- Analyst takeaway:
- Betweenness captures *potential control*, not always *actual control*.

Closeness Centrality Breaks in Real Networks

- **Failure Mode 3: Closeness in Fragmented or Directed Networks**
- Closeness assumes:
  - A connected network
  - Symmetric reachability
- Real networks violate both:
  - Disconnected communities (echo chambers)
  - Directed networks (Twitter, command hierarchies)
  - Gatekeepers block information asymmetrically
- Result:
  - Some nodes appear mathematically unreachable
  - Metric becomes unstable or meaningless
- Analyst takeaway:
- Closeness works in theory. Fragmentation breaks it in practice.

Metrics Are Snapshot Illusions

- **Failure Mode 4: Temporal Instability**
- Centrality is often computed on **static graphs**
- But real networks are:
  - Temporal (edges appear/disappear)
  - Event-driven (crises (emergency) restructure connectivity)
  - Strategic (actors adapt to avoid detection)
- Example:
  - A node central today may be peripheral tomorrow after one meeting or one arrest
- Analyst takeaway:
- Structure is dynamic. Static centrality is often a historical artifact.

Dangerous Failure: Metric Worship

- **Failure Mode 5: Confusing Numbers with Understanding**
- Common mistake:
  - “Highest centrality = most important node”
- But:
  - Metrics are **lenses**, not truths
  - Different questions require different measures
- Real analyst workflow:
  - Start with question
  - Choose metric
  - Interpret within context
  - Validate against domain knowledge
- Analyst takeaway:
- Centrality scores do not replace judgment. They discipline it.

Exercise: You Are the Network Analyst

- Scenario: Analyzing a covert coordination network.
- Each node = individual, Each edge = trusted communication channel
- Objective: Disrupt the network with minimal intervention.
- You are allowed to remove only 2 nodes.

Cont.

- Three candidates emerge:
- Node A: Highest degree (talks to many)
- Node B: Highest betweenness (connects two clusters)
- Node C: Moderate centrality but highly trusted leader inside one cluster
- Questions: Which two would you remove?
- What outcome are you optimizing for?
- Fragmentation?
- Confusion?
- Loss of coordination?
- What could go wrong with your decision?

Option 1: Remove A (high degree) + B (high betweenness)

- **Strengths**
- Structural impact:
  - Removing B likely **fragments the network** into disconnected components
  - Removing A reduces **local broadcast efficiency**
- High immediate observable disruption:
  - Fewer messages
  - Longer paths
  - Reduced coordination across clusters
- **Weaknesses**
- Strategic failure risk:
  - C (trusted leader) remains intact
  - Leadership survives; network can **reconstitute**
- Networks often regenerate bridges:
  - A new broker may quickly emerge

Likely outcome - Over-optimizes for **structure**, ignores **social legitimacy**

Option 2: Remove A (high degree) + C (trusted leader)

- **Strengths**
- Removes: The loudest communicator (A) and The authority figure (C)
- Social impact:
  - Loss of legitimacy
  - Loss of morale
  - Potential internal confusion
- Especially powerful in:
  - Ideological groups, Startups, Political movements
- **Weaknesses**
- Structural problem remains:
  - B still bridges clusters
  - Network topology still intact
- Coordination can persist:
  - A new leader may emerge through B's connectivity
- **Likely outcome**
- Psychological destabilization without guaranteed structural collapse.
- //A *sociologically intelligent but structurally naïve* answer.

Option 3: Remove B (high betweenness) + C (trusted leader)

- **Strengths**
- Structural + social collapse:
  - B removal fragments the network
  - C removal eliminates leadership coherence
- Results in:
  - Isolated clusters
  - No central authority
  - No bridge for coordination
- This is closest to:
  - How counterterrorism targeting works
  - How corporate crisis collapse happens
  - How decentralized movements die
- **Weaknesses**
- Leaves A (high degree) active:
  - Noise continues
  - High chatter remains
- Can give illusion of activity despite collapse of strategy
- **Likely outcome -** Network appears active but becomes strategically incoherent and fragmented.

Summary

- This is the difference between computing networks and understanding networks

Example - Terror Cell

- Suppose we are analyzing a terror network graph.
  Each node = individual
  Each edge = communication / coordination / logistics connection.
- Should we arrest the person with highest degree?
- Will that always collapse the network?
- The answer:
- Sometimes yes. Often no.
- It depends on **structure**, not just degree.

Example cont.

- High-degree nodes in such networks may correspond to: Recruiters, Propaganda disseminators, Logistics coordinators, Funding intermediaries, Communication relays
- But: Some high-degree nodes are deliberately designed as **decoys**.
- Networks often evolve to **hide their true backbone**.
- Thus: Leaders reduce their visible degree
- Cells compartmentalize, Communication is intentionally sparse
- So: Degree centrality is necessary, but insufficient.

Analogy

- Imagine a drug cartel or terror network:
- Couriers talk to many people
- Foot soldiers interact widely
- The strategist speaks to only 2–3 trusted lieutenants
- Who has higher degree? The courier.
- Who is more critical? The strategist.

Disruption as a Network Problem

- The goal of disruption is not to arrest individuals, but to degrade network functionality.
- Network-level goals:
- Increase path lengths
- Break connectivity between modules
- Reduce coordination capacity
- Fragment the giant component
- Increase mistrust across edges

Effective Disruption…

- High-degree nodes = visible activity
- Real power often lies in:
- Low-degree brokers
- Trusted intermediaries
- Structural bridges
- In networks, the loudest node is rarely the most powerful.
- True power lies in structural position, not visibility.
- Effective disruption requires:
- Degree + Betweenness + Community structure

Revisiting Terror-Cell Framing

- Suppose we map a covert network:
- Nodes = individuals, Edges = communication / coordination links
- We observe:
- Several tightly-knit small groups (cells)
- Very few connections between groups
- A handful of individuals who connect those groups
- Those connectors may:
- Have **low degree,** Stay **invisible socially,** Avoid attention deliberately
- Yet structurally: They are disproportionately powerful.

Node Type	Degree	Betweenness	Structural Role
Recruiter	High	Low	Noisy but replaceable
Foot soldier	Medium	Low	Local function
Courier between cells	Low	High	Structural bridge
Strategist	Low	Very high	Network backbone

Moving forward

- Community detection
- Information diffusion
- Trust and influence
- Betweenness → predicts community bridges
- Centrality → influences diffusion models
- Structure → determines trust propagation
- The most powerful actor is often the one you never notice — until the structure collapses.

Key Takeaway

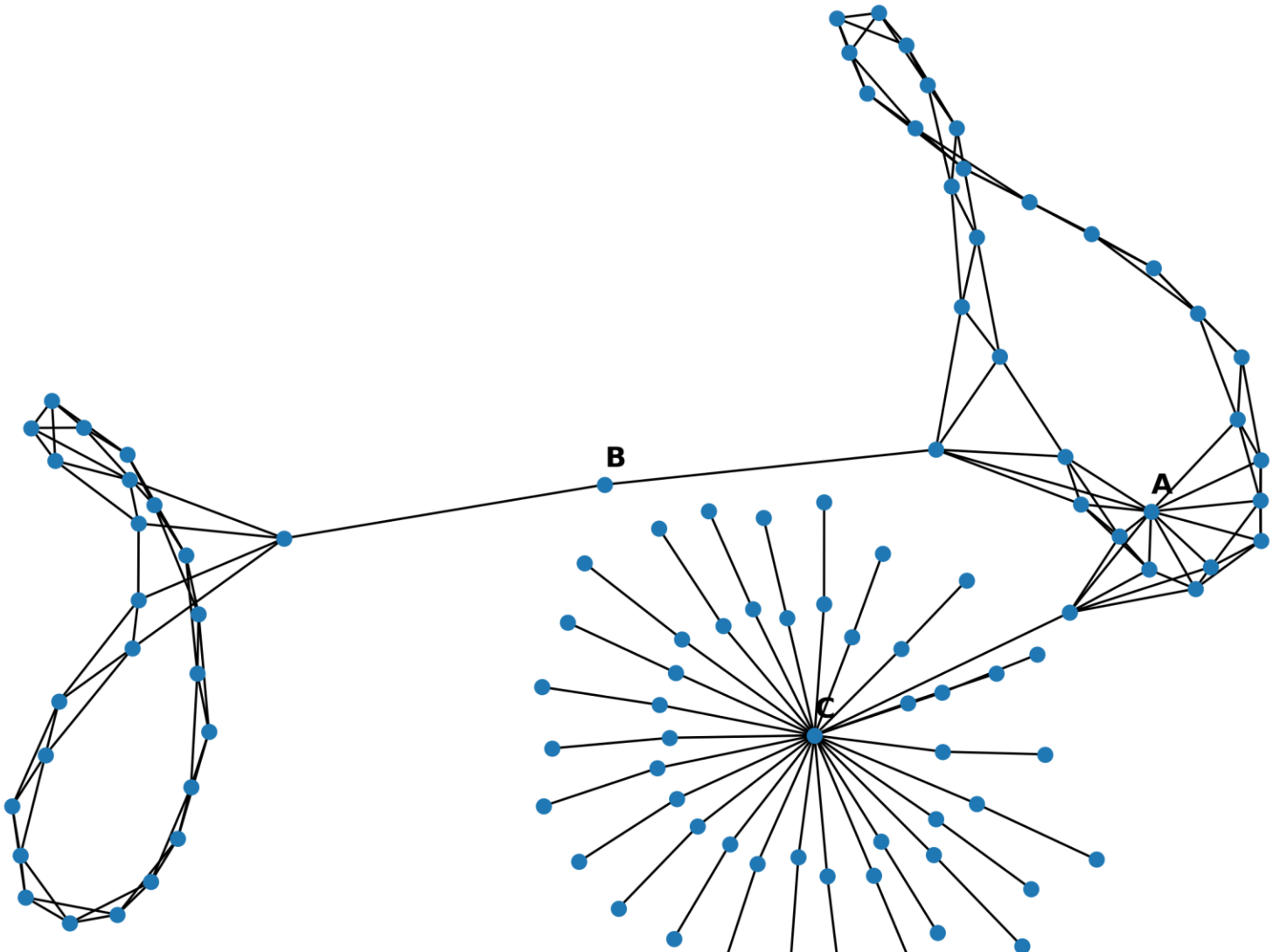
- *Social Network Analysis* (SNA) investigates complex interactions between people, groups and organizations – e.g. events that trigger the rapid spread of emerging stories in social media
- Outcomes emerge from structure
- In complex networks, the most powerful actor is often the one you never notice—
  until the structure collapses when they are removed.
- SNA reveals invisible systems

Reflection Task

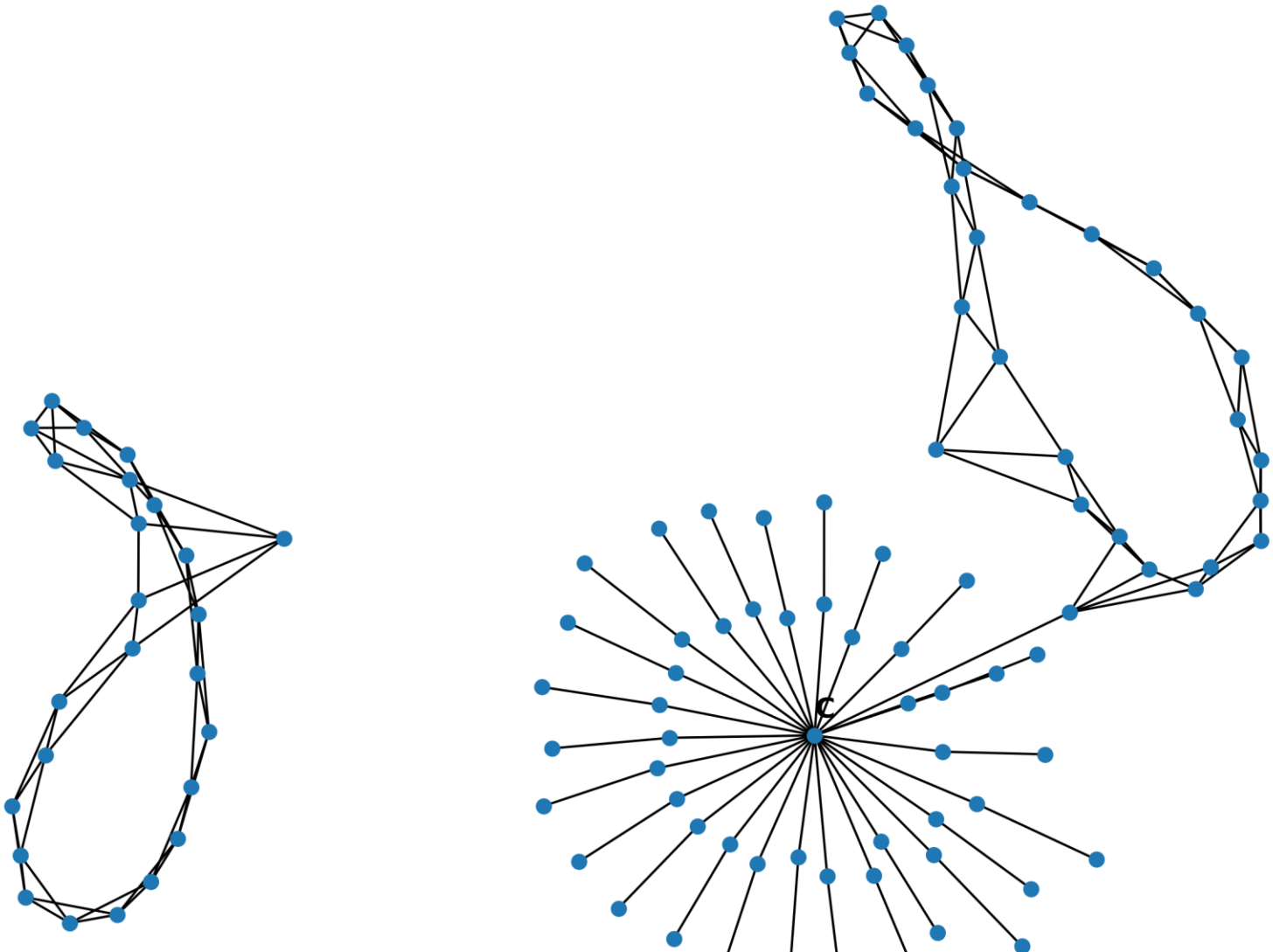
- Observe a real network
- Identify hubs and bridges
- Reflect on structure
- Example – subsequent slides

Constraint-Driven Quantitative SNA Example

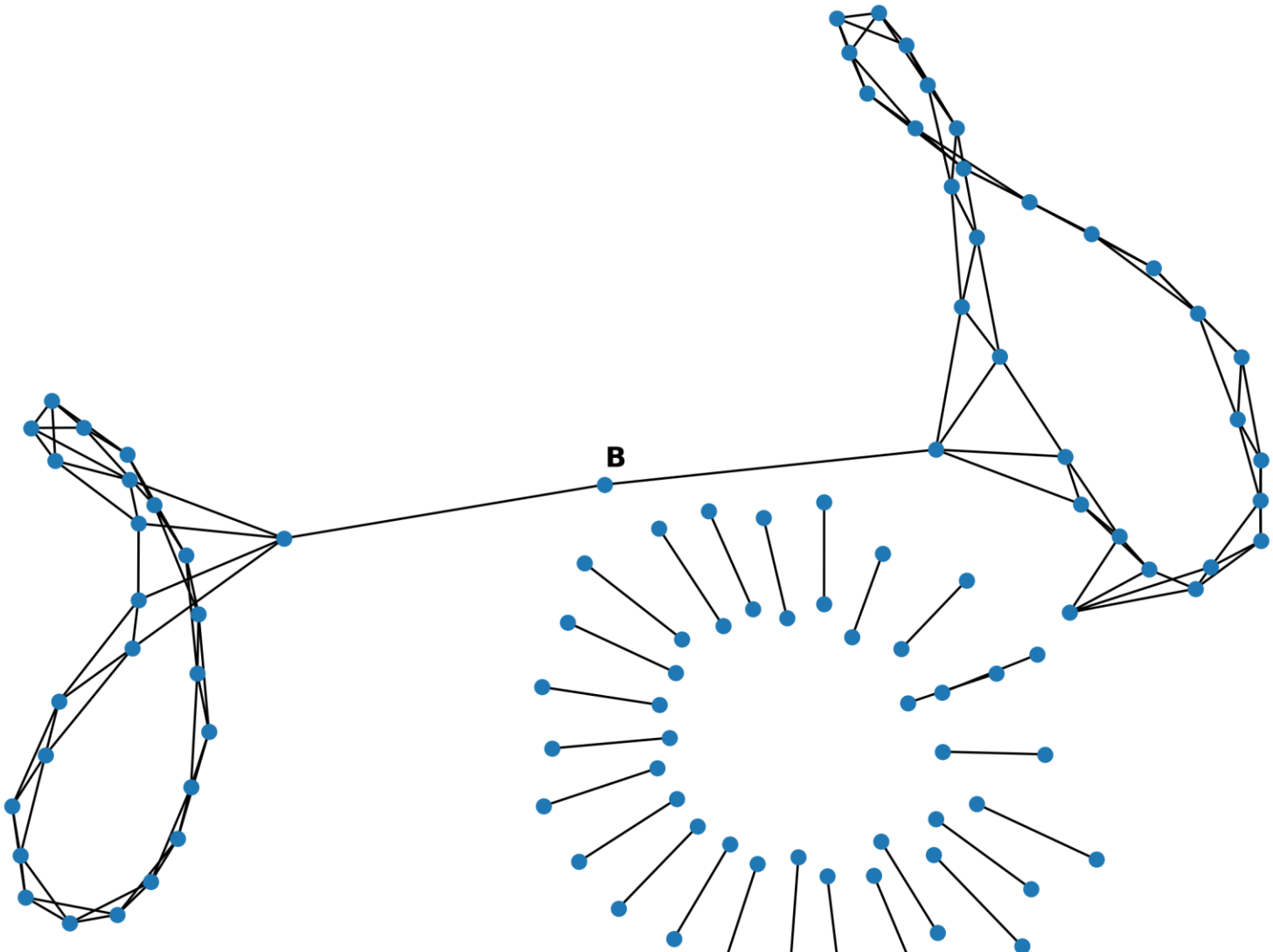
Original Network



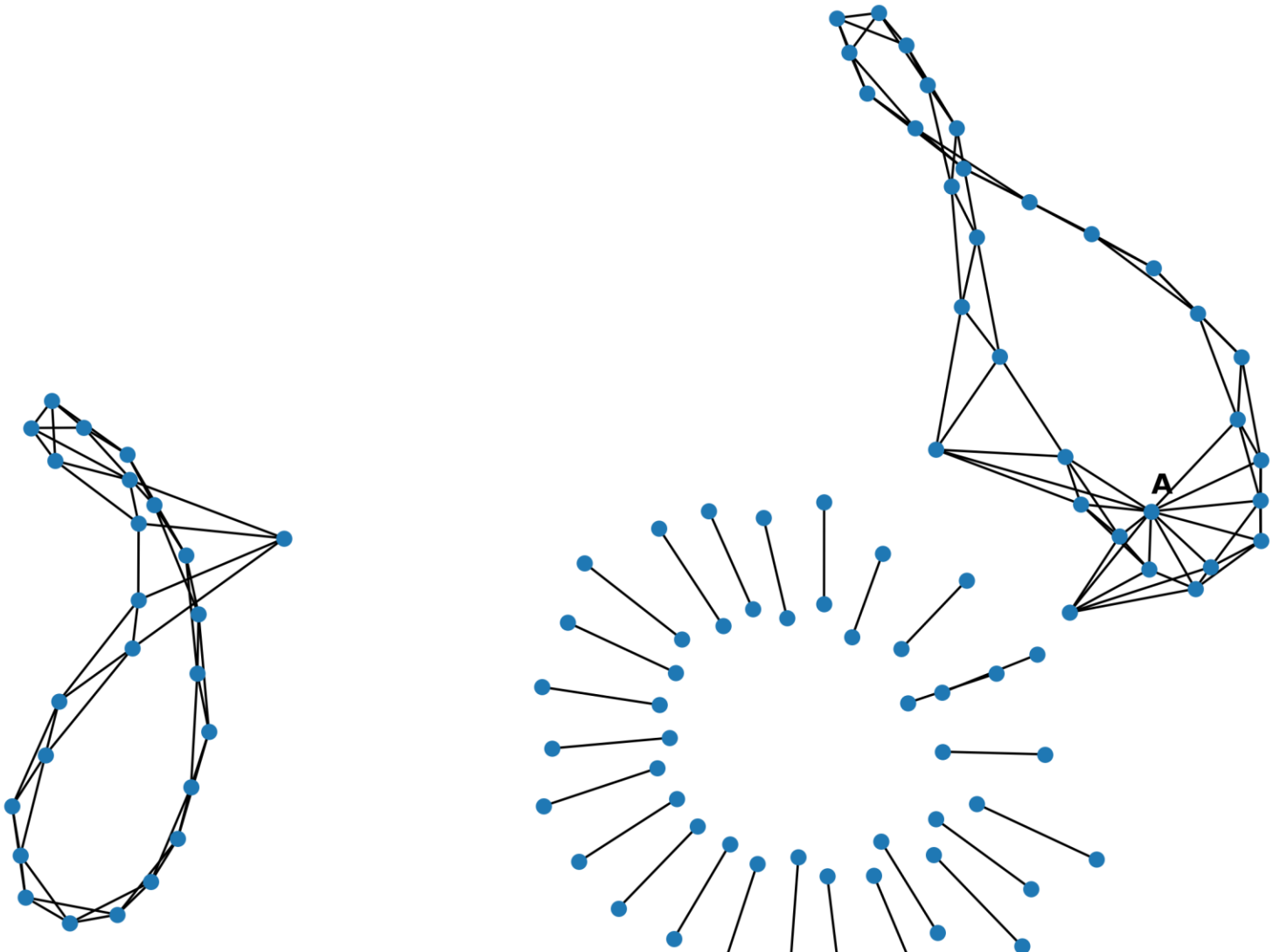
Remove A + B



Remove A + C



Remove B + C (Worst Outcome)



Quantitative Evidence

Strategy	# Components	Largest Component
Original	1	108
A+B	2	81
A+C	26	56
B+C	27	31

Why?

- B is the only bridge between two large, internally robust communities
- C controls many fragile peripheral cells (satellites)
- A has high visibility (degree) but no structural leverage
- Removing B + C simultaneously attacks:
- Global connectivity (via B)
- Local fragmentation (via C)
- **Why B + C dominates (structural explanation)**
- B = inter-community articulation
- C = controller of many dependent subgraphs
- A = high degree but redundant

End

- Earlier we debated which two nodes to remove.
- Now we built a quantitative model and tested it.
- The data gives us a deeper truth: There is no universally optimal strategy — only strategies optimal for different goals.
- SNA is decision science, not arithmetic